

A student performed an experiment to determine the energy content of a sample of potato chips. The chips were blended and then freeze-dried to remove moisture from the sample. Freeze-drying, or lyophilisation, is the preferred dehydration process for food calorimetry due to its minimal impact on nutritional content. Lyophilisation removes water from frozen food samples at low pressure through sublimation. The water is later collected in a cooled condensation chamber to prevent damage to any electric components.

After freeze-drying, the sample was ground into a dry homogenous powder. A 1-g sample of the powder was placed in a bomb cell, an enclosed steel-walled container, and pressurized to 25 atm with pure oxygen to ensure complete combustion. The bomb cell was submerged in 500 g of water at room temperature (25°C) in an enclosed and completely thermally isolated calorimetry device. The sample was combusted using an electric fuse. The temperature of the surrounding water was measured with a glass thermometer every 30 s for several minutes (Figure 1).

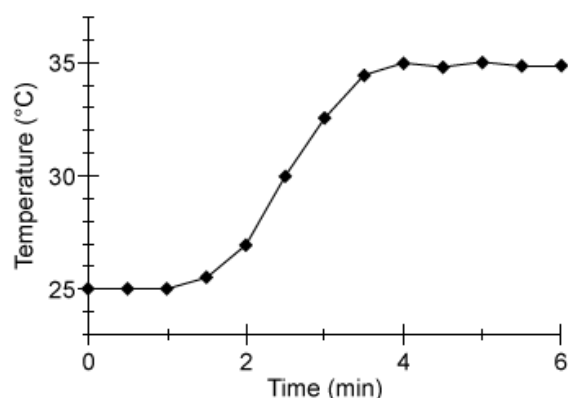


Figure 1 Temperature of the water in the bomb calorimeter

The energy content of food is given in units of dietary calories (Cal), which is equal to 4.185 kJ, or 1,000 gram calories (cal). A gram calorie is the amount of energy required to increase 1 g of water by 1°C.

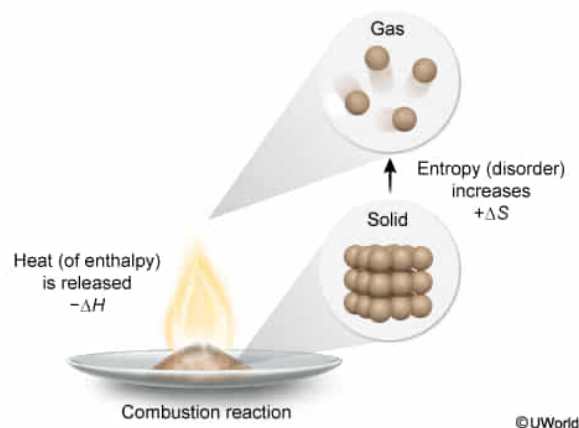
For the combustion reaction of the sample into CO_2 (g) and H_2O (g) in the bomb cell:

- ☐ A. ΔH and ΔS are both positive. [20%]
- ☐ B. ΔH and ΔS are both negative. [5%]
- ☒ C. ΔH is negative and ΔS is positive. [64%]
- ☐ D. ΔH is positive and ΔS is negative. [8%]

Correct

64% answered correctly

Explanation:



Enthalpy (H) is the thermodynamic quantity that measures the energy stored in a chemical bond, and ΔH is the net amount of energy (heat) released or absorbed from the formation and destruction of chemical bonds in a reaction. Energy is released as heat if the enthalpy of the products is lower than the enthalpy of the reactants, and ΔH is negative (exothermic). Because the temperature of the surroundings (water) increases in the combustion of the sample, **heat is released** and therefore **ΔH is negative**.

Entropy (S) is the thermodynamic quantity that measures the disorder of a substance (or system). Solids have relatively low entropy because the structure of the molecules is ordered and their motion is more constrained. Gases have relatively high entropy because the molecules are disordered and their motion is less constrained. Because the gaseous products of the reaction are more disordered than the solid powdered sample, **disorder increases** and therefore **ΔS is positive**.

(Choices A and D) Although the enthalpy (heat) of the surroundings increases, ΔH describes the reaction. ΔH would be positive (endothermic) if energy is absorbed in a chemical reaction.

(Choice B) ΔS would be negative if disorder decreases (order increases).

Educational objective:

Enthalpy (H) is a measure of bond energy, and a change in the enthalpy of a reaction results in the release or absorption of heat. Entropy (S) is a measure of disorder, and it is highest in gases and lowest in solids due to their respective molecular structures. In the combustion of a solid to a gas, heat is released (ΔH is negative) and the products are more disordered (ΔS is positive).

Time spent: 0 sec
Subject:
Physics

QID: 401164
System:
Thermodynamics

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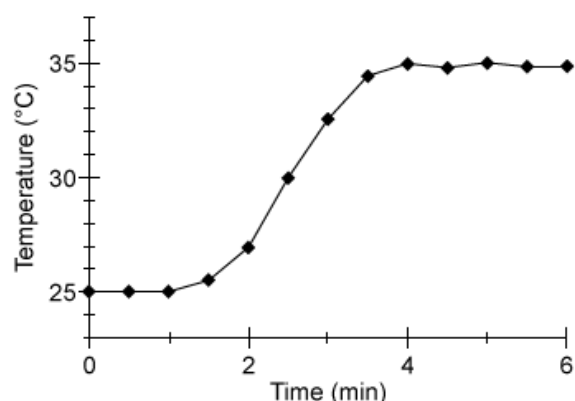


Figure 1 Temperature of the water in the bomb calorimeter

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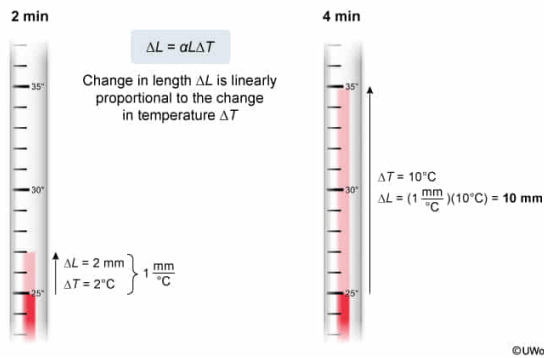
After the experiment had progressed for 2 min, the liquid in the thermometer rose 2 mm. How high will the liquid rise from the 25°C mark after the experiment had progressed for 4 min?

- ☐ A. 4 mm [12%]
- ☐ B. 6 mm [5%]
- ☐ C. 8 mm [12%]
- ✓ ☒ D. 10 mm [69%]

Correct

70% answered correctly

Explanation:



The **thermal expansion** of the length L or volume V of a substance is linearly proportional to its change in temperature ΔT :

$$\Delta L = \alpha L \Delta T$$

$$\Delta V = \alpha_V V \Delta T$$

where α or α_V is the coefficient of thermal expansion specific to the substance.

Although the liquid in the thermometer will expand volumetrically, it is confined in a column, and therefore its expansion causes its **height to increase linearly with temperature**. According to the question, the change in the height of the liquid ΔL is 2 mm after the first 2 min, which corresponds to a temperature change ΔT of 2°C (Figure 1):

$$\alpha L = \frac{\Delta L}{\Delta T} = \frac{2 \text{ mm}}{2^\circ\text{C}} = 1 \frac{\text{mm}}{^\circ\text{C}}$$

The height of the liquid increases by 1 mm for every increase in 1°C . This ratio is used to predict ΔL after 4 min, which corresponds to $\Delta T = 10^\circ\text{C}$ (Figure 1).

$$\Delta L = \left(1 \frac{\text{mm}}{^\circ\text{C}}\right)(\Delta T)$$

$$\Delta L = \left(1 \frac{\text{mm}}{^\circ\text{C}}\right)(10^\circ\text{C}) = 10 \text{ mm}$$

The height of the liquid in the thermometer will increase by **10 mm** after 4 min.

(Choice A) The thermal expansion of a substance is linearly proportional to the change in temperature, not the change in time.

(Choice C) The question asks for the change in height at 4 min (35°C) from the 25°C mark (0 min). If the change in temperature $\Delta T = 8^\circ\text{C}$ from 2 min to 4 min is incorrectly used, the calculated change in height will be 8 mm.

Educational objective:

The expansion of a substance is linearly proportional to its change in temperature, and it is described by its coefficient of thermal expansion. Given a change in temperature

and its corresponding change in length, the linear relationship can be applied to predict the change in length at other temperatures.

Time spent:0 sec

Subject:

Physics

QID:401165

System:

Thermodynamics

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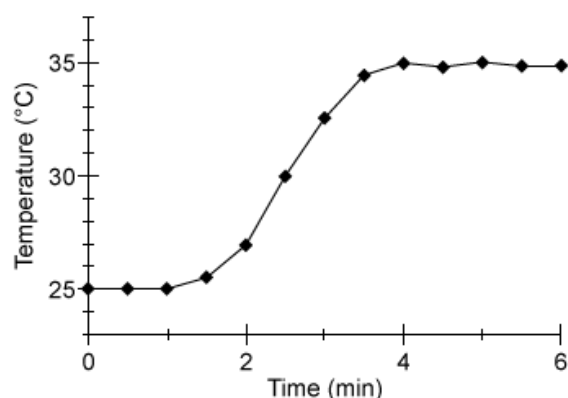


Figure 1 Temperature of the water in the bomb calorimeter

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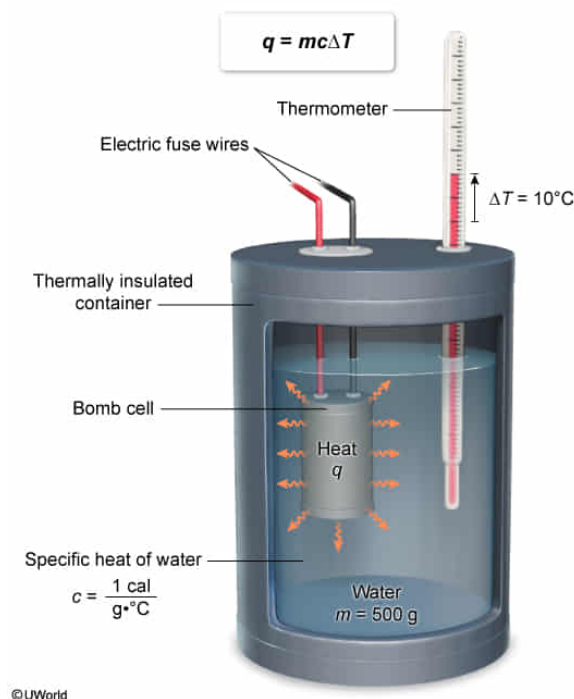
Approximately how much heat in kJ was released from the combustion of the 1-g sample?

- ☐ A. 5 kJ [11%]
- ✓ ☒ B. 20 kJ [37%]
- ☐ C. 40 kJ [48%]
- ☐ D. 70 kJ [1%]

Correct

38% answered correctly

Explanation:



Calorimetry is a technique used to measure heat transfer from changes in temperature. The amount of **heat** q required to change the temperature of a substance is the product of its mass m , specific heat c (the amount of heat required to increase 1 g of a substance by 1°C), and the change in temperature ΔT :

$$q = mc \Delta T$$

In the experiment described in the passage, the heat released from the combustion reaction is absorbed by the surrounding water ($m = 500 \text{ g}$). The description of a gram calorie (cal) is the definition of water's specific heat: $c = 1 \text{ cal}/(\text{g} \cdot ^\circ\text{C})$. The maximum temperature change $\Delta T = 10^\circ\text{C}$ (Figure 1) is due to the total heat released from the combustion reaction. The heat q released in cal is the product of these values (**Choice A**):

$$q = (500 \text{ g}) \left(1 \frac{\text{cal}}{\text{g} \cdot ^\circ\text{C}} \right) (10^\circ\text{C}) = 5,000 \text{ cal}$$

The value of 5,000 cal of heat energy is converted to kJ using the conversion provided in the passage: $1,000 \text{ cal} = 4.185 \text{ kJ} \approx 4.2 \text{ kJ}$ (rounded to two significant digits).

$$q = (5,000 \text{ cal}) \left(\frac{4.2 \text{ kJ}}{1,000 \text{ calories}} \right) = 21 \text{ kJ}$$

This value is closest to **20 kJ**.

(Choice C) The mass of the water is used in the calorimetry equation because it absorbs the heat from the combustion reaction. If the mass of the sample (1 g) and the incorrect conversion $1 \text{ cal} \approx 4.2 \text{ kJ}$ are used, q is calculated as 42 kJ.

(Choice D) The amount of heat transferred depends on the change in temperature ΔT , not the final temperature reached. If the final temperature of the water (35°C) is

incorrectly used, q is calculated as 70 kJ.

Educational objective:

Calorimetry is a technique used to measure heat transfer from changes in temperature. The specific heat of a substance is the amount of heat required to increase 1 g of a substance by 1°C. The change in heat q is the product of the mass m , the specific heat c , and the change in temperature ΔT : $q = mc\Delta T$.

Time spent:0 sec

Subject:
Physics

QID:401166

System:
Thermodynamics

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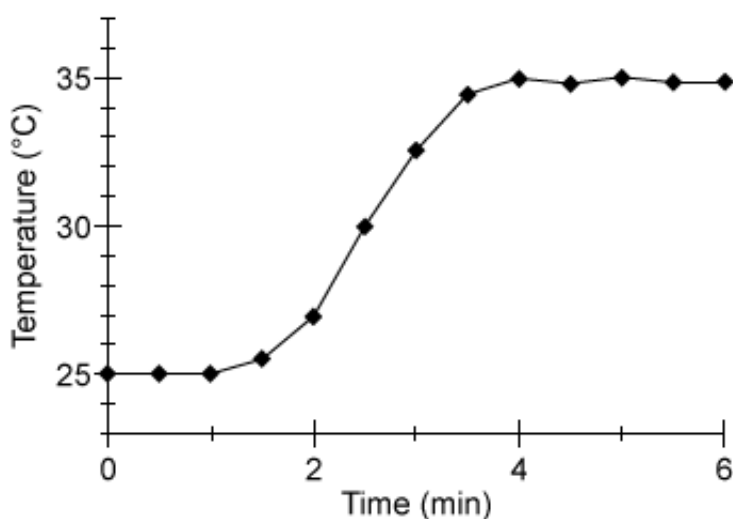


Figure 1 Temperature of the water in the bomb calorimeter

The energy content of food is given in units of dietary calories (Cal), which is equal to 4.185 kJ, or 1,000 gram calories (cal). A

gram calorie is the amount of energy required to increase 1 g of water by 1°C.

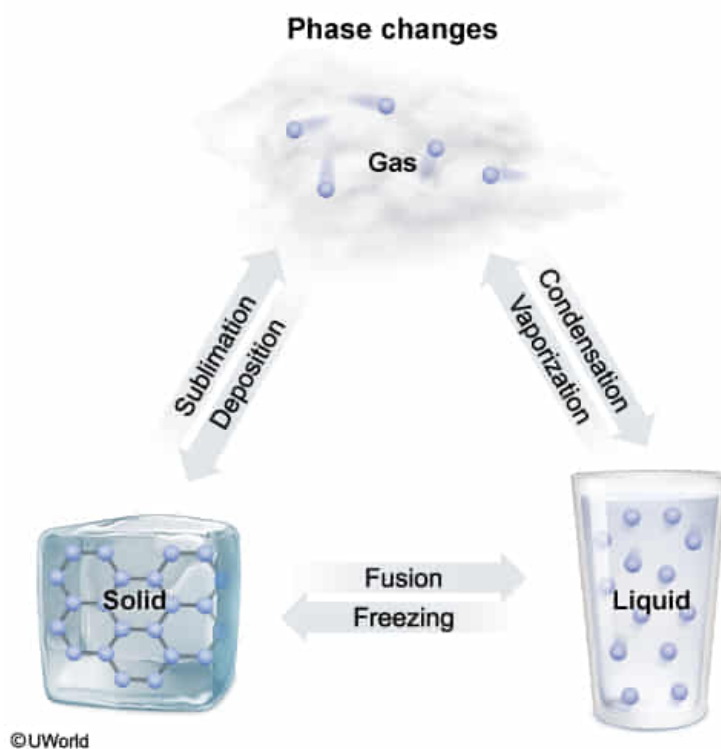
What is the order of the phase changes of water throughout the lyophilisation process after freezing?

- ✓ ☒ A. Solid to gas to liquid [75%]
☐ B. Solid to gas to solid [10%]
☐ C. Solid to liquid to gas [10%]
☐ D. Solid to liquid to solid [2%]

Correct

76% answered correctly

Explanation:



The phases of matter describe the uniform physical state of the matter's molecules, and include solid, liquid, and gas:

- Molecules in a solid are tightly packed in an ordered structure and retain a fixed shape.

- Molecules in a liquid move past each other (but stay close together) and assume the shape of their container at a fixed volume.
- Molecules in a gas move past each other, can be well separated, and assume the shape and volume of their container.

The phase of matter depends on the temperature and pressure of its environment, and therefore changes in pressure and/or temperature can cause phase changes.

According to the passage, the lyophilisation process begins with a frozen sample, so the water in the sample starts out in its solid phase. Afterward, the frozen water is **sublimated at low pressures**, and **sublimation** is the phase change of a solid directly to a **gas**. In the last step, the sublimated water vapor (gas) is collected in a **condensation** chamber, and condensation is the phase change of a gas to a **liquid**.

(Choices B and D) The phase change from a gas to a solid is deposition, and the change from a liquid to a solid is freezing. There is no indication that the water returned to a solid at the end of the lyophilisation process.

(Choice C) The phase changes from solid to liquid and liquid to gas are melting and vaporization, respectively. The sublimation of water is the direct transition from its solid phase to its gas phase without entering the intermediary liquid phase.

Educational objective:

The phases of matter describe the physical state of the matter's molecules, and include solid, liquid, and gas. Sublimation is the phase change from solid to gas, and condensation is the phase change from gas to liquid.

Time spent:0 sec

Subject:

Physics

QID:401167

System:

Thermodynamics

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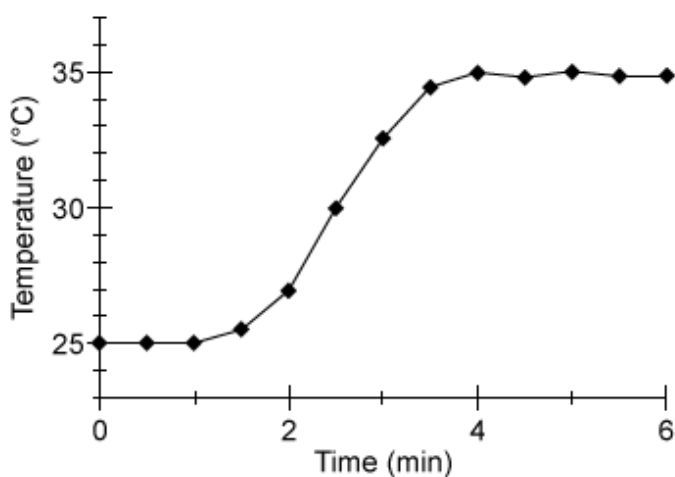


Figure 1 Temperature of the water in the bomb calorimeter

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If the student repeated the experiment by replacing the water in the calorimetry device with an ice bath at 0°C, how would the experimental

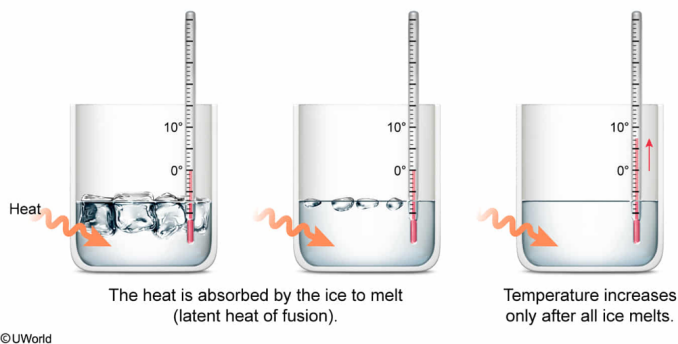
results differ?

- ☐ A. The temperature of the water would increase to 10°C. [17%]
- ✓ ☒ B. The temperature of the water would begin to increase at a later time. [57%]
- ☐ C. The heat released when ice melts increases the measured temperature. [11%]
- ☐ D. The amount of heat released from the combustion reaction would increase. [13%]

Correct

57% answered correctly

Explanation:



At the melting (or freezing) point of a substance, the substance can exist in either its solid or liquid phase. Because the **freezing/melting point of water** is 0°C, the ice bath will remain as a mixture of solid ice and liquid water unless heat is added to or removed from the mixture. The **latent heat of fusion** (melting) is the amount of heat (energy) required to convert a solid at its melting point temperature to its liquid phase by breaking the bonds between the molecules in the solid phase.

The heat released from the combustion reaction will first go toward the latent heat of fusion to **melt the ice** into liquid water. The temperature of the water will be constant at 0°C as long as ice is still present and **will not start to increase** until after all the ice melts. Therefore, the **temperature** of the water would begin to **increase at a later time** (compared to the original experiment) when enough heat is transferred from the combustion reaction to first melt the ice (latent heat of fusion).

(Choice A) Because some of the heat released goes toward the latent heat of fusion of the ice to melt it, less heat will be available to increase the temperature of the water. Therefore, the magnitude of the temperature increase would be less than the 10°C temperature change in the original experiment.

(Choice C) Ice absorbs heat when it melts. Conversely, liquid water releases heat equal to the latent heat of fusion when it freezes.

(Choice D) The amount of heat released from the combustion would be unchanged because it depends on the mass of the reactants (powdered sample), which is unchanged.

Educational objective:

The phase transition from a solid to a liquid requires heat (energy) to break the bonds between molecules; this energy is the latent heat of fusion. When heat is added to a mixture of ice and water at 0°C , the heat will first go toward melting the ice before raising the temperature of the water.

Time spent:0 sec

Subject:
Physics

QID:401168

System:
Thermodynamics

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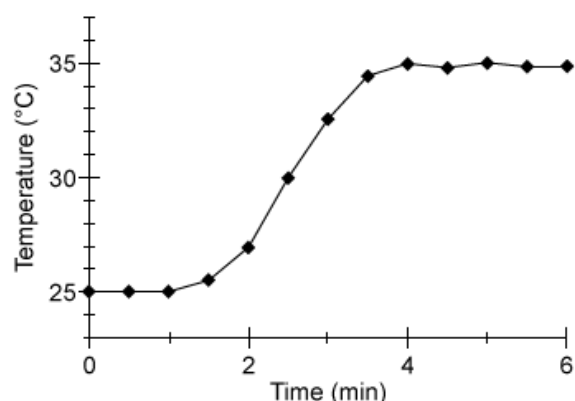
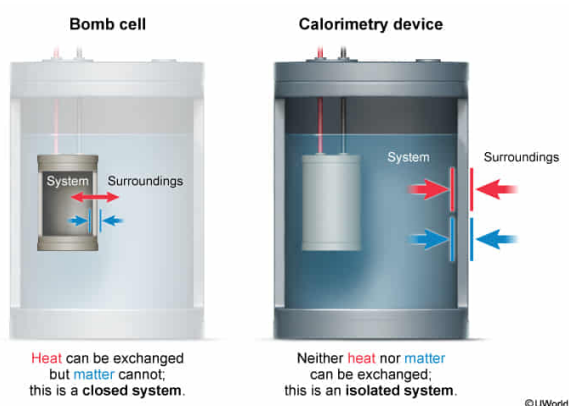


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The bomb cell and the calorimetry device used in the experiment are what types of thermodynamic systems?

- ☐ A. Both are closed systems. [10%]
- ☐ B. Both are isolated systems. [7%]
- ☒ C. The bomb cell is a closed system, and the calorimetry device is an isolated system. [59%]
- ☐ D. The bomb cell is an isolated system, and the calorimetry device is a closed system. [22%]

Explanation:

In thermodynamics, the "system" is the physically enclosed space being studied that is separated from the rest of the universe, known as the "surroundings." There are three main types of **thermodynamic systems**, which are characterized by the type of barrier between the system and its surroundings:

- An open system allows heat and matter to be exchanged with the surroundings.
- A closed system allows heat but not matter to be exchanged with the surroundings.
- An isolated system does not allow heat or matter to be exchanged.

According to the passage, the **bomb cell** houses the combustion reaction, and its surroundings include the water in the calorimetry device. Because the temperature of the surrounding water increased during the combustion reaction, **heat can be exchanged** between the bomb cell and its surroundings. Because the container is enclosed and does not allow for the exchange of matter, the bomb cell is an example of a **closed system**.

The **calorimetry device** houses the bomb cell and the water it is submerged in, and its surroundings are the surface it rests on and the nearby air. Because the enclosed walls of the device are completely thermally isolated (poor conductors of heat), **neither heat nor matter** can be exchanged with its surroundings, and the device is an example of an **isolated system**.

(Choice A) The calorimetry device is an isolated system, not a closed system.

(Choices B and D) The bomb cell is a closed system, not an isolated system.

Educational objective:

An open system allows heat and matter to be exchanged with the surroundings. A closed system allows heat but not matter to be exchanged with the surroundings. An isolated system does not allow heat or matter to be exchanged with the surroundings.